

CAGAYAN DE ORO, Philippines

WMO: 987480

Lat: 8.484N

Lon: 124.647E

Elev: 6

StdP: 101.25

Time Zone: 8.00 (E08)

Period: 86-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.1	22.8	18.3	13.2	23.8	19.4	14.2	24.4	5.8	27.7	5.0	30.3	0.9	220	0.316

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	8.3	34.6	27.5	34.1	27.4	33.6	27.3	28.8	33.1	28.4	32.8	28.1	32.5	2.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	23.6	32.3	27.2	23.0	32.1	26.8	22.4	31.8	93.8	33.2	92.0	33.0	90.3	32.7	32.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.8	3.7	3.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	27.0	27.4	27.8	28.6	29.4	28.6	28.4	28.7	28.5	28.3	28.0	27.5
	DBStd	1.17	1.11	1.05	0.99	0.96	0.94	0.99	0.94	1.05	0.91	0.82	1.00	1.03
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6643	528	487	553	557	603	558	570	580	556	567	541	544
	CDD18.3	3601	269	254	294	307	344	308	312	322	306	309	291	286
	CDH23.3	39048	2535	2502	3172	3626	4317	3399	3317	3628	3296	3363	3032	2861
	CDH26.7	15754	818	884	1253	1570	2042	1397	1336	1547	1351	1354	1177	1027
(14) Wind	WSAvg	1.1	1.1	1.0	1.2	1.1	1.1	0.9	1.0	1.0	1.0	1.0	1.1	1.1
(15) Precipitation	PrecAvg	1667	119	86	51	32	104	215	201	228	220	178	138	95
	PrecMax	2041	452	181	159	79	232	357	413	331	373	316	372	241
	PrecMin	1101	0	4	2	0	12	62	45	109	57	58	23	11
	PrecStd	257	128	61	44	24	72	80	105	69	91	75	97	74
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	33.3	34.1	34.2	35.5	34.7	34.5	35.5	34.6	34.1	34.0	33.4
		MCWB	26.5	26.2	26.6	27.8	27.4	27.8	27.2	27.9	27.5	27.5	26.9	27.2
	2%	DB	32.2	32.5	33.1	33.6	34.6	33.7	33.6	34.1	33.7	33.3	33.1	32.5
		MCWB	26.3	26.1	26.5	27.4	27.5	27.3	27.5	27.4	27.3	27.4	26.9	27.0
	5%	DB	31.5	31.9	32.5	33.0	34.0	33.1	32.9	33.3	33.0	32.7	32.5	31.9
		MCWB	26.1	26.1	26.5	27.1	27.5	27.2	27.2	27.2	27.2	27.3	26.8	26.8
	10%	DB	30.6	31.1	31.7	32.5	33.3	32.4	32.2	32.6	32.3	32.1	31.8	31.1
		MCWB	25.9	26.0	26.5	27.0	27.3	27.0	26.9	27.0	26.9	27.1	26.7	26.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.9	27.7	28.5	29.0	29.2	29.1	29.0	28.9	28.7	28.7	28.5	28.2
		MCDB	31.7	31.8	32.3	33.4	33.7	33.5	33.1	33.6	33.3	32.8	32.4	31.8
	2%	WB	27.2	27.2	27.7	28.2	28.6	28.3	28.3	28.2	28.1	28.1	27.9	27.6
		MCDB	30.9	31.3	31.8	32.6	33.3	32.9	32.6	33.0	32.7	32.4	32.0	31.5
	5%	WB	26.7	26.7	27.2	27.7	28.1	27.8	27.7	27.7	27.7	27.6	27.4	27.1
		MCDB	30.4	30.9	31.5	32.0	32.9	32.3	32.2	32.5	32.2	32.0	31.5	31.2
	10%	WB	26.1	26.2	26.7	27.3	27.6	27.2	27.2	27.2	27.2	27.2	26.9	26.7
		MCDB	29.9	30.4	31.0	31.6	32.5	31.6	31.5	32.0	31.6	31.6	31.0	30.6
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.5	8.1	8.2	8.3	8.0	8.1	8.3	8.3	8.1	8.1	7.8
		MCDBR	8.7	8.8	9.2	8.9	8.9	8.7	9.0	9.1	9.1	8.9	9.0	8.7
		MCWBR	4.9	4.8	5.3	5.0	4.5	4.5	4.9	4.8	4.9	5.0	5.0	5.0
	5% WB	MCDBR	7.9	8.1	8.4	8.4	8.5	8.2	8.5	8.5	8.7	8.4	8.2	8.4
		MCWBR	5.0	4.9	5.4	5.2	4.9	4.8	5.2	5.0	5.1	5.1	5.1	5.2
(40) Clear-Sky Solar Irradiance	taub		0.408	0.404	0.406	0.417	0.426	0.417	0.416	0.438	0.437	0.444	0.414	0.408
	taud		2.490	2.482	2.484	2.450	2.440	2.471	2.470	2.393	2.390	2.368	2.472	2.487
	Ebn at Noon		905	918	915	892	866	864	868	862	874	869	892	897
	Edh at Noon		110	113	114	117	115	110	111	122	124	125	111	108
(44) All-Sky Solar Radiation	RadAvg		4.11	4.84	5.46	5.86	5.45	5.03	4.97	5.17	5.13	4.92	4.59	4.30
	RadStd		0.54	0.54	0.42	0.33	0.33	0.24	0.27	0.34	0.34	0.32	0.39	0.38

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	+0.22	N/A	N/A	N/A	+0.44	+0.33	+0.31	N/A	N/A	N/A	N/A

Nomenclature: See separate page